



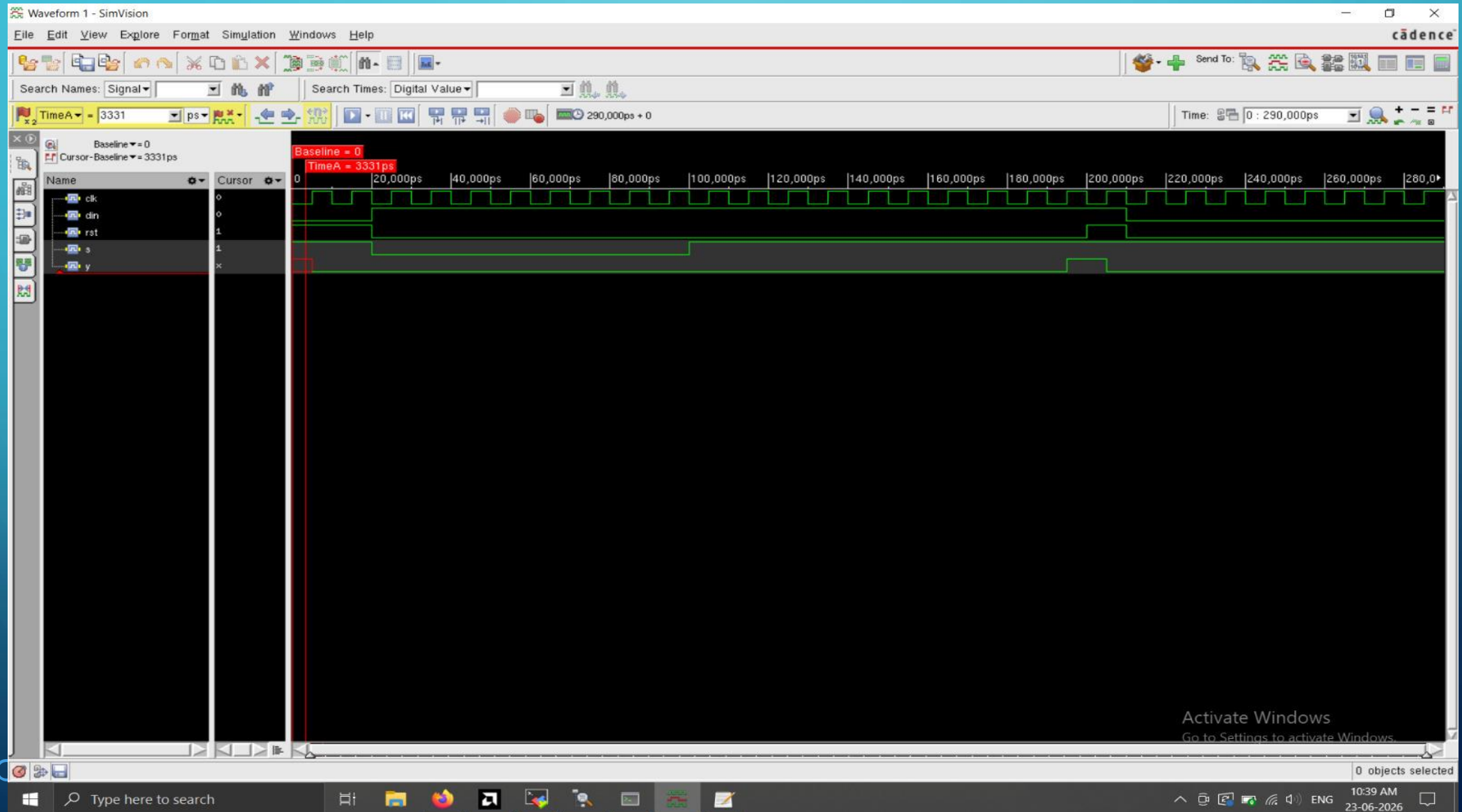
DOOR LOCK USING PASSKEY

- **OBJECTIVE: A CHIP THAT UNLOCKS A DOOR ONLY WHEN THE CORRECT 8-DIGIT CODE IS ENTERED. IF THE USER ENTERS THE SEQUENCE CORRECTLY, THE DOOR UNLOCKS. IF A WRONG DIGIT IS ENTERED, THE SYSTEM RESETS AND STARTS OVER.**

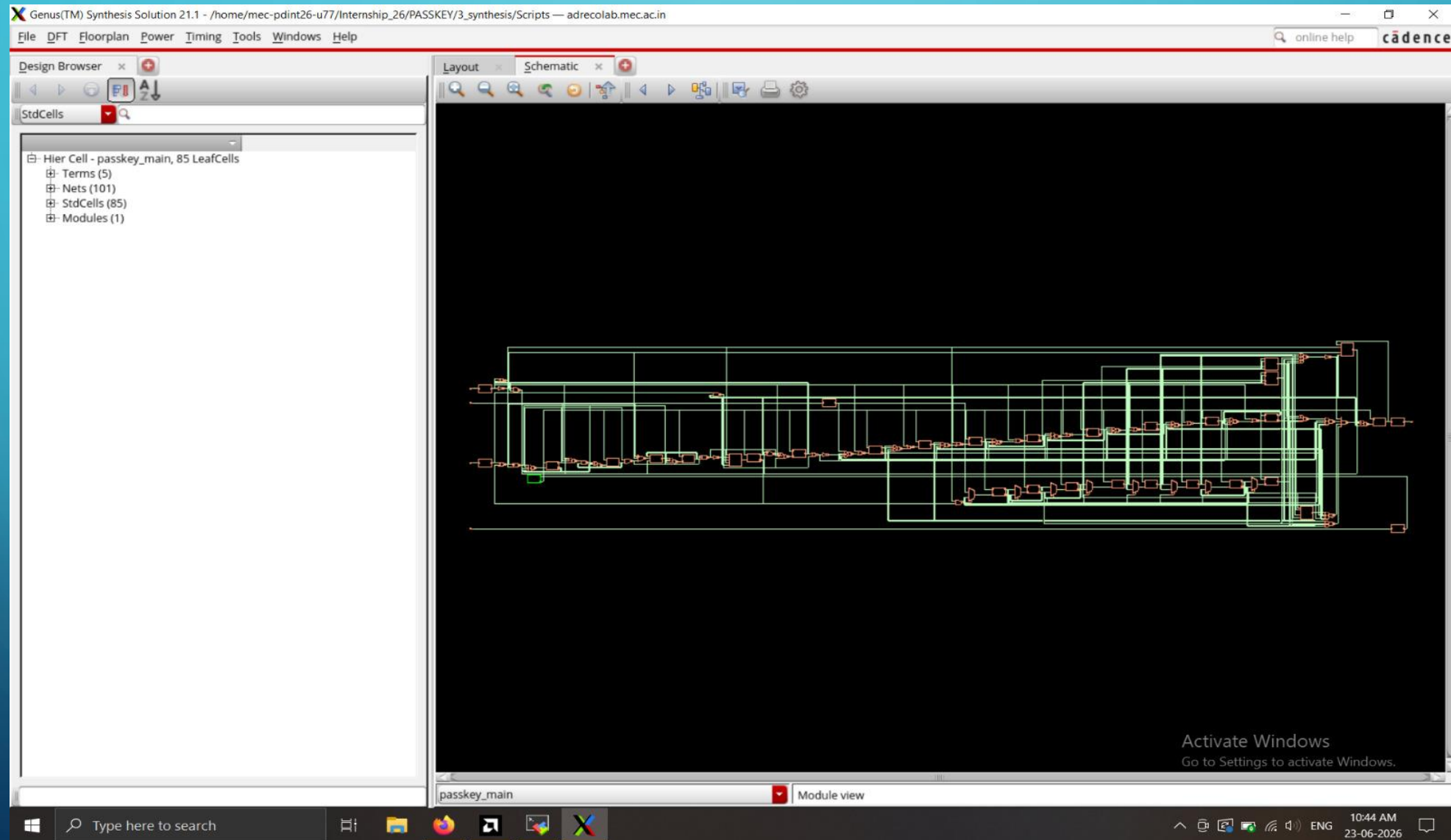
BLOCK DIAGRAM



SIMULATION



SYNTHESIS



TIMING

192.168.8.23 (mec-pdint26-u77)

Open 

design_timing.rpt
~/Internship_26/PASSKEY/3_synthesis/report_files

```
1 =====
2 Generated by:      Genus(TM) Synthesis Solution 21.14-s082_1
3 Generated on:      Jun 23 2026  11:32:30 am
4 Module:           passkey_main
5 Operating conditions:  ts118cio150_max (worst_case_tree)
6 Wireload mode:    enclosed
7 Area mode:        timing library
8 =====
9
10
11 Path 1: MET (17963 ps) Late External Delay Assertion at pin y_pad
12   Group: R20
13   Startpoint: (R) y_reg/CP
14   Clock: (R) passkey_clk
15   Endpoint: (R) y_pad
16   Clock: (R) vir_main_clk
17
18           Capture      Launch
19   Clock Edge:+ 30000      0
20   Src Latency:+ 0        3000
21   Net Latency:+ 0 (I)    0 (I)
22   Arrival:= 30000      3000
23
24   Output Delay:- 4000
25   Required Time:= 26000
26   Launch Clock:- 3000
27   Data Path:- 5037
28   Slack:= 17963
29
30 Exceptions/Constraints:
31   output_delay      4000      constraints.sdc_line_56
32
33 #-----
34 # Timing Point  Flags  Arc  Edge  Cell  Fanout  Load Trans Delay Arrival Instance
35 #              (fF)  (ps) (ps) (ps) (ps)  Location
36 #-----
37 y_reg/CP        -      -    R  (arrival)  22      -  1000    0    3000  (-,-)
38 y_reg/Q         -      CP->Q R  dfnrq1    2 186.7  2144  1453  4453  (-,-)
39 pc3o05_1/PAD    -      I->PAD R  pc3o05    1 50.0   845   3584  8037  (-,-)
40 y_pad          <<<      -    R  (port)    -      -    -    0    8037  (-,-)
41 #-----
42
```

Windows  Type here to search 

POWER

192.168.8.23 (mec-pdint26-u77)

Open 

design_power.rpt
~/Internship_26/PASSKEY/3_synthesis/report_files

1 Instance: /passkey_main
2 Power Unit: W
3 PDB Frames: /stim#0/frame#0

Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	1.79007e-08	4.48369e-05	6.93061e-06	5.17854e-05	4.71%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	1.06186e-08	7.36932e-06	6.96184e-06	1.43418e-05	1.30%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	3.09504e-04	3.09504e-04	28.14%
pad	2.00618e-06	6.07832e-04	1.14573e-04	7.24412e-04	65.85%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	2.03470e-06	6.60038e-04	4.37970e-04	1.10004e-03	100.00%
Percentage	0.18%	60.00%	39.81%	100.00%	100.00%

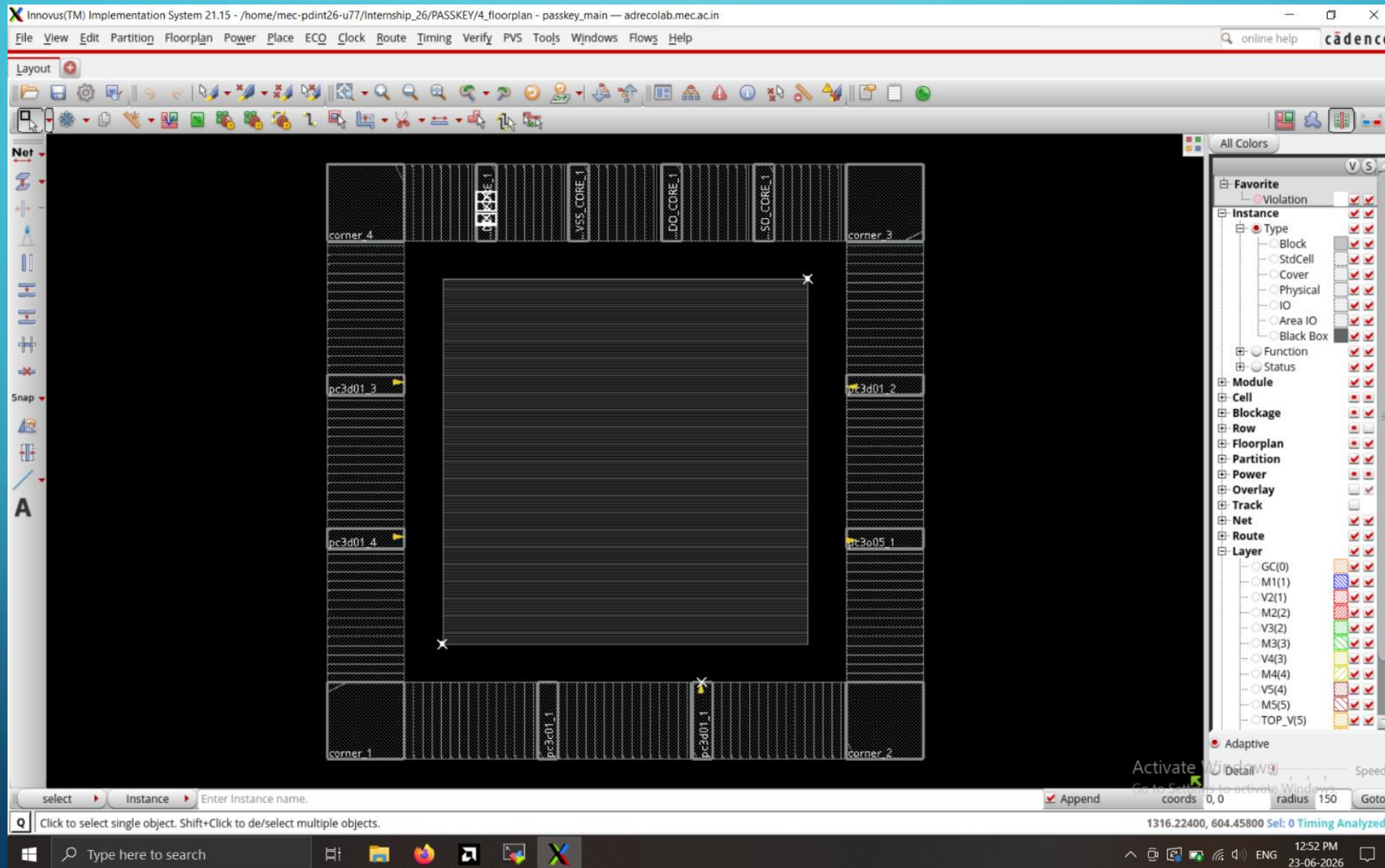
Windows Taskbar: Type here to search, File Explorer, Google Chrome, Microsoft Edge, VS Code, Notepad

AREA

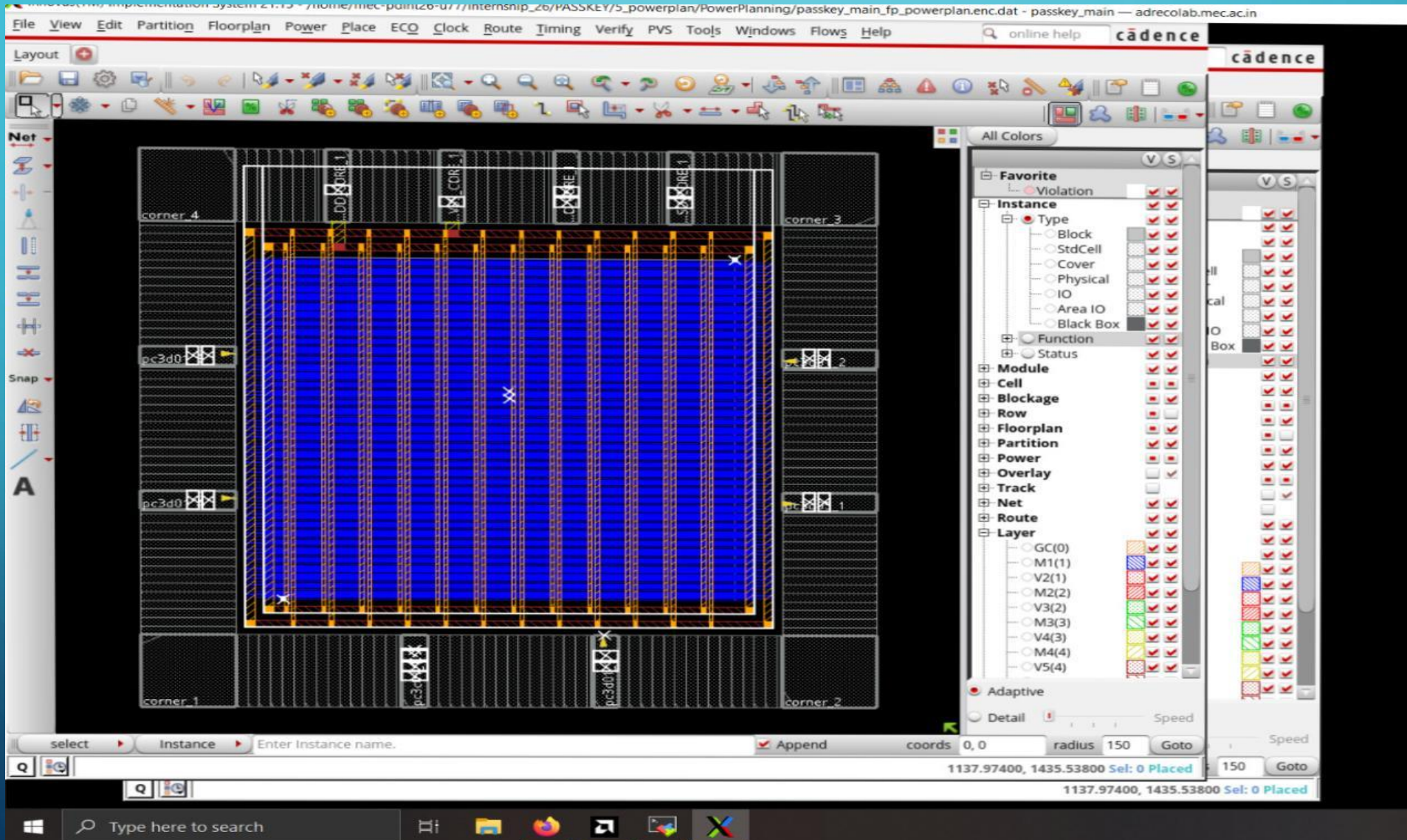
```
Innovus(TM) Implementation System 21.15 - /home/mec-pdint26-u77/Internship_26/PASSKEY/8_routing/Routing/passkey_main_PostMetalfillRoute.enc.dat - passkey_
Open  [icon] design_area.rpt
~/Internship_26/PASSKEY/3_synthesis/report_

1 =====
2 Generated by:      Genus(TM) Synthesis Solution 21.14-s082_1
3 Generated on:      Jun 23 2026  11:32:30 am
4 Module:            passkey_main
5 Technology libraries:  tsl18fs120_scl_ss 1
6                    tsl18ciol50_max 1.0
7                    tsl18fs120_scl_ff 1
8                    tsl18ciol50_min 1.0
9 Operating conditions:  tsl18ciol50_max (worst_case_tree)
10 Wireload mode:      enclosed
11 Area mode:          timing library
12 =====
13
14 Instance  Module  Cell Count  Cell Area  Net Area  Total Area  Wireload
15 -----
16 passkey_main      88  10174.760   39.830   10214.590  16000 (S)
17 (S) = wireload was automatically selected
```

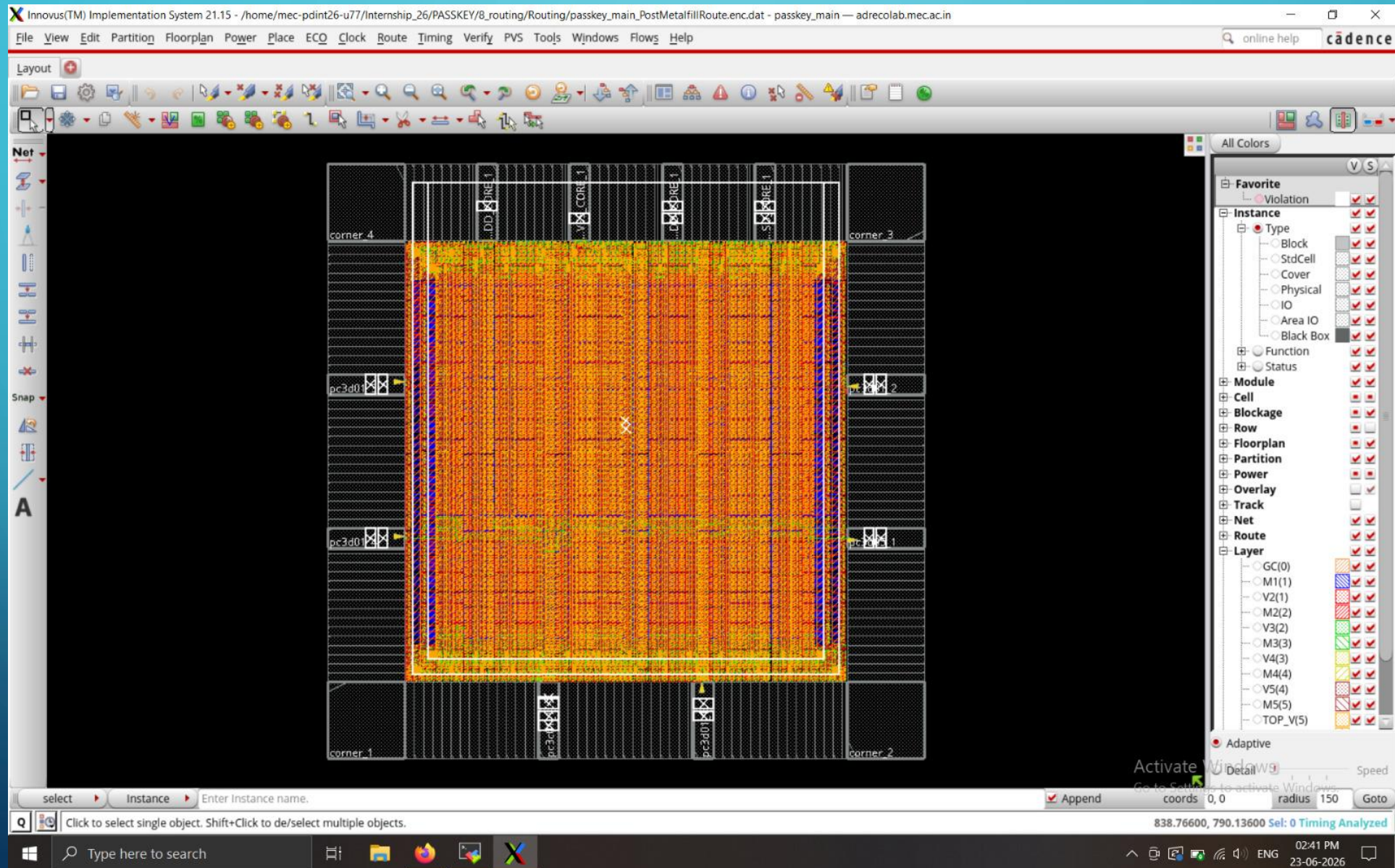
FLOOR- PLANNING



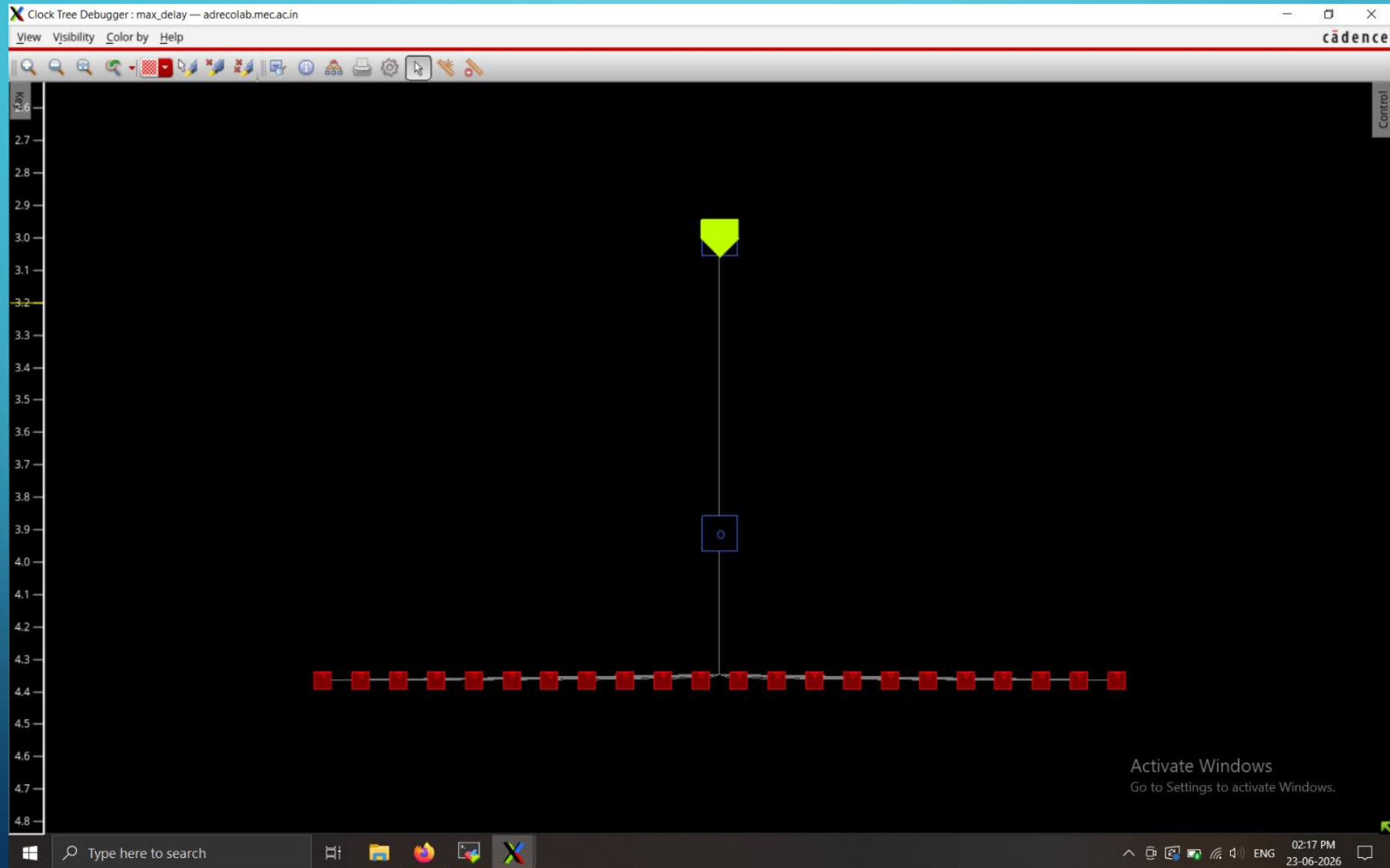
POWER-PLANNING



PLACEMENT



CLOCK



ROUTING

